

18. (3 points)

Given the following information for a retrospectively rated book of business as of December 31, 2015:

Policy Effective Year	Policy Effective Quarter	Ultimate Loss	Losses Reported at Prior Retro Adjustment
2011	1	\$92,500	\$92,500
2011	2	57,000	57,000
2011	3	125,000	125,000
2011	4	80,000	78,000
2012	1	64,000	57,000
2012	2	37,500	30,000
2012	3	60,000	55,000
2012	4	65,000	60,000
2013	1	55,000	50,000
2013	2	40,000	33,000
2013	3	70,000	52,500
2013	4	60,000	45,000
2014	1	50,000	0
2014	2	40,000	0
2014	3	65,000	0
2014	4	45,000	0
2015	1	25,000	0
2015	2	20,000	0
2015	3	30,000	0
2015	4	5,000	0

Retro Adjustment Period	Selected PDL Ratio	Percent Loss Emerged	Policy Period	Premiums Booked from Prior Adjustment	Premiums Booked as of December 31, 2015
First	1.75	78.5%	2011	\$450,000	\$452,000
Second	0.70	10%	2012	335,000	337,000
Third	0.55	7%	2013	330,000	335,000
Fourth	0.45	4%	2014-2015	0	425,000
Subsequent	0.00	0.5%			

Calculate the premium asset as of December 31, 2015.

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EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 18

TOTAL POINT VALUE: 3

LEARNING OBJECTIVE: A14: Forecast premium reserves.

SAMPLE ANSWERS

Sample Answer 1

$$CPDLD = \frac{\sum PDLD \times \% \text{ Loss Emerged}}{\sum \% \text{ Loss Emerged}}$$

CPDLD1 = 1.5
CPDLD2 = 0.588
CPDLD3 = 0.491
CPDLD4 = 0.4

Expected Future Loss = Ult – Loss Reported as of Prior

AY	Expected Future Loss
11	2,000
12	24,500
13	44,500
14 & 15	280,000

Expected Future Prem = Expected Future Loss × CPDLD
Prem Asset = Expected Future Prem + Prior Booked – Current Booked

AY	CPDLD	Future Premium	Premium Asset
11	0.4	800	-1,200
12	0.491	1,2029.5	10,029.5
13	0.588	26,166	21,166
14 & 15	1.5	420,000	-5,000
24,995.5			

= 330,000 + 26,166 – 335,000

Sample Answer 2

$$CPDLD = \text{sumproduct of future incurred loss dev and PDLD} / \sum$$

Retro Adj	CPDLD
1	1.5
2	.588
3	.4913
4	.4

Year	Next Retro is	Est Future Prem
11	4	800
12	3	12,037
13	2	26,166
14	1	420,000
15	1	

Est Future Prem = (Ult loss – loss reported at prior) × CPDLD of next adj

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

Prem asset = prem from prior adj + est future prem – premium booked now

Prem asset = 1,574,003 – 1,549,000 = 25,003

EXAMINER'S REPORT

Candidates were expected to know most calculations and how to interpret the item, with the understanding that it requires a long calculation with many subtle details.

The most common errors candidates made were simple calculation errors or calculator errors.

Also, a few candidates:

- Did not calculate expected future loss correctly
- Applied the CPDLD ratios to the wrong periods
- Used PDLD ratios instead of CPDLD ratios
- For the Premium asset formula, added current booked premium to expected future premium and subtracted prior booked premium (whereas they should have added prior booked premium to expected future premium and subtracted current booked premium)

Most candidates did very well (either received full credit or lost credit only for minor calculation errors).